DNA Barcoding for Exploration of Taxonomic Diversity and Ecological Management of Ant Species in Washington, D.C.

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ABSTRACT

Ant populations serve a vital role in our local ecosystems by recycling dead matter, enriching the soil with nutrients, and contributing to the flow of the food pyramid, maintaining a stable ecological balance. Taxonomic diversity is the measure of the number of species per genus and family within a community and can provide valuable insight into necessary conservation efforts to be taken for these creatures. Ant samples were collected on Howard University's campus, extracted for their DNA, sequenced, and finally analyzed through DNA Subway. Online ant databases, such as AntWeb and AntMaps, were then used to assess the populations of similar subfamilies, genera, and species within the D.C. region in which our samples were collected. Results demonstrated that the two predominant subfamilies in Washington, D.C. are Formicidae and Mycriminae. The Strumigenys and Temnothorax genera of the Mycriminae subfamily have the greatest number of native species present (9 and 7 respectively). Similarly, the Lasius and Formica genera of the Formicidae subfamily are the most abundant, both with 8 native species. The study of these predominant subfamilies, genera, and species can also help us to further understand environmental dynamics such as nutrient cycling, the presence of invasive species, habitat requirements, etc. Surveillance will aid in conserving these species that may not be as abundant non-locally and conserving at-risk species that may share the same habitat.

INTRODUCTION

In this study, we investigate the ant population dynamics in Washington, D.C., recognizing their pivotal role in shaping taxonomic diversity and ecosystem dynamics. Ants, as prominent inhabitants of urban environments, have significant influence over ecosystem processes. Understanding their population dynamics is crucial for assessing the health and stability of urban ecosystems. Ant samples were collected across various habitats within the city by Howard University, employing standardized methods to ensure comprehensive coverage. By analyzing these samples, we aim to explain the taxonomic composition of ant communities and evaluate their potential impacts on ecosystem functioning. This research contributes to our understanding of urban ecology and underscores the importance of preserving biodiversity in urban landscapes for the well-being of both humans and the environment.

MATERIALS & METHODS

1) Collection:

For the ant collection and documentation, we used bait traps to gather ants and established a habitat profile while observing their behavior.

2) Extraction:

Following this, we utilized Chelex DNA extraction method, where ant tissues were ground in Chelex within a microcentrifuge tube, then centrifuged to extract DNA for genetic analysis. Subsequently, the entire ant was immersed in the Chelex solution for a minimum of 2 minutes before a portion of the liquid was collected, labeled, and stored on ice.

3) Gel Electrophoresis & PCR:

The DNA obtained was mixed with loading dye and stored separately. Gel electrophoresis was employed to separate DNA molecules by size using electric current through an agarose gel matrix. Polymerase chain reaction (PCR) was conducted to amplify a specific segment of ant DNA, particularly targeting the COI gene, leading to the successful amplification of high-quality DNA products suitable for subsequent analysis. DNA sequencing was performed to identify the bases (A, T, C, G) within each DNA fragment, ensuring accurate results through quality control measures.

4) Data Analysis and Species Search:

Bioinformatics analysis was then carried out using DNALC's online pipeline to analyze the DNA sequences obtained. Finally, using the data collected from our sample's phylogenetic tree, we conducted a region search on AntWeb and AntMaps to compile the number of species native to Washington D.C. per common genus and subfamily.

RESULTS

Figure 1. Native species per subfamily in D.C.

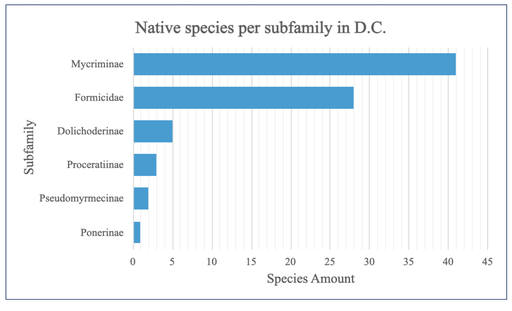


Figure 1. Pie chart showing the number of species present in the D.C. region according to the six primary subfamilies. The subfamilies with the most abundant species amount are the Mycriminae (41) and Formicidae (28).

Figure 2. Species per genus of the Mycriminae Subfamily

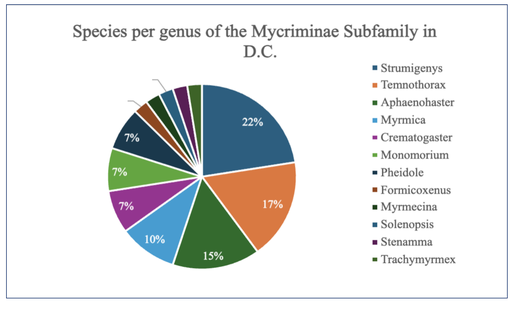


Figure 2. Pie chart showing the number of species within each of the twelve primary genera of the most abundant subfamily in the D.C. region: Mycriminae. The two genera with the greatest number of native species present in D.C. are the Strumigenys (22%) and Temnothorax (17%).

RESULTS

Figure 3. Species per genus of the Formicidae Subfamily

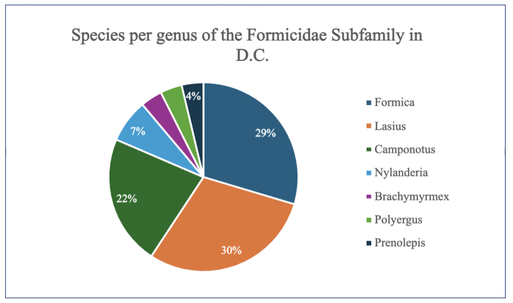


Figure 3. Pie chart showing the number of species within each of the seven primary genera of the second most abundant subfamily in the D.C. region: Formicidae. The two genera with the greatest number of native species present in D.C. are the Lasius (30%) and Formica (29%).

DISCUSSION

Through the collection, sequencing, and analysis process of nearby ant populations, we were able to identify our sample as a Pheidole Bicarinata species of the Mycriminae subfamily. Further research on this subfamily led to the discovery that it is the second most abundant within the region behind the Formicidae. Fifty-nine percent of the Formicidae subfamily population are species of the genera Formica and Lasius. Thirty-nine percent of the Mycriminae subfamily population are species of the genera Strumigenys and Temnothorax. This study has also revealed the subfamilies and genera that the D.C. region is devoid of, such as the Ponirae subfamily with only one species or the Pheidole genus—like that of our ant sample—with only three species. Knowledge of the thriving and declining species in our local region can be used towards future research on the different environmental and behavioral factors affecting their composition.

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